Credit Name: CSE 2140
Assignment Name: Chapter

<u>Question #1</u>

a) 4 legal identifiers names
   i) int validExample = 0;
   ii) boolean running = true;
   iii) String word = “word”;
   iv) int number = 7;
b) 4 illegal identifiers and why
   i) int incorrect Example; - an identifier cannot have a space
   ii) String word = word; - the string word is assigned to needs “” around it
   iii) boolean running = 3; - booleans can only be assigned to true or false
   iv) int num = “3”; - putting “” around the number makes it a string (even if the number inside is a number).

<u>Question #2</u>

a) Creating variable numBeads (2 Statements)
   i) int numBeads;
   ii) numBeads = 5;
b) Creating variable numBeads (1 Statements)
   i) int numBeads = 5;

<u>Question #3</u>

a) Final value of yourNumber
   i) Int myNumber = 5;
   ii) Int yourNumber = 4;
   iii) myNumber = yourNumber * 2; // myNumber = 8
   iv) yourNumber = myNumber + 5;
   v) yourNumber equals 13 when the program ends
b) Final value of yourNumber
   i) Int myNumber;
   ii) Int yourNumber = 4;
   iii) myNumber = yourNumber + 7; // myNumber = 11
   iv) yourNumber = myNumber;
   v) yourNumber equals 11 when the program ends

<u>Question #4</u>

Appropriate data types

a) Num basketballs
   i) Int
b) Price of basketball
   i) double
c) Number of player on basketball team

    i) int
- d) Average age of player on basketball team
    - i) double
- e) Whether a basketball player has a jersey or not
    - i) boolean
- f) First initial of the basketball players first name
    - i) String

Question #5

- a) Different between primitive and abstract data type
    - i) Primitive data types store simpler values, stuff like integers, true/false, decimals
    - ii) Abstract data types store a thing or a reference to something earlier in the code, stuff like words or lists
- b) difference between class and object
    - i) A class is a blueprint for different information about something, for example if a athlete was a class t could include their sport, height, strength, or position
    - ii) An object is what is what that information makes, for example you could have tim, who plays football, is ‘6”2, benches 225lbs and is a quarterback

Question #11

int j = 5;
double k = 1.6;
int y;
double z;

- a) y = j * k;
    - i) Should be converted to int
    - ii) y = int (j* k);
- b) z = j * k;
    - i) The adjustment is automatic
- c) z = k * k;
    - i) Remains double the entire time
- d) j = k;
    - i) Needs to be converted to int
    - ii) J = int k
- e) k = j;
    - i) Automatic conversion
- f) y = j + 3;
    - i) Both are integers